

# Full Stack Open Part 0: The Verdict

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The purpose of Part 0 is to teach the **evolution of web development** so that you understand exactly *what problem React solves* before you write a single line of React code.

Here is what the three exercises are meant to drill into your brain:

## 1. Exercise 0.4 (The Traditional Web)

- **The Lesson:** This is how the web used to work (and how older frameworks like PHP or Ruby on Rails still often work).
- **The Takeaway:** The server is the boss. The browser is "dumb." Every time you do an action (like submit a form), the server forces the browser to throw away the entire page and download *everything* again (HTML, CSS, JS, JSON).
- **The Problem:** It's slow, inefficient, and creates a "flickering" user experience because the page has to constantly reload.

## 2. Exercise 0.5 (The SPA Initial Load)

- **The Lesson:** This is the foundation of modern web apps (React, Vue, Angular).
- **The Takeaway:** Instead of the server sending a fully built HTML page, the server sends a mostly empty HTML file and a **massive JavaScript file** (`spa.js`).
- **The Shift in Power:** The browser is now "smart." The JavaScript file takes over. From this point forward, JavaScript handles all the logic, fetches the raw data (JSON), and builds the UI directly in the browser.

## 3. Exercise 0.6 (The SPA Action)

- **The Lesson:** This is the magic of Single Page Applications (SPAs).
- **The Takeaway:** When you submit a form, the page **does not reload**. JavaScript intercepts the action (e.g. `preventDefault()`), updates the screen instantly using the data it already has in memory, and quietly sends a single `POST` request to the server in the background to save it.
- **The Solution:** It's incredibly fast. The user gets instant feedback, and we only send a tiny piece of JSON data over the network instead of downloading the whole website again.

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## The Final Verdict: Why did they make you do this?

If you jump straight into React without understanding this, React just feels like a confusing way to write HTML in JavaScript.

But because you did these diagrams, you now understand that **React is just a tool designed to do exactly what you saw in 0.5 and 0.6.**

React exists to take control of the browser, stop page reloads, manage data (state) in memory, and update the screen instantly without needing the server to send new HTML.

You now understand the *why*. You are officially ready to learn the *how*.